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Tutorial Week 4

COMP3670 / COMP6670: Introduction to Machine Learning

The four questions are adapted from Abu-Mostafa, Magdon-Ismail & Lin (2012), *Learning from Data*, Exercises 2.3, 2.6, 5.2, 5.3 and Problem 2.3.

Hoeffding's inequality, as we shall use it. If a hypothesis h is **fixed before** a sample of size N is looked at, then with probability at least $1-\delta$

$$\left|E_{\text{in}}(h) - E_{\text{out}}(h)\right| \;\leq\; \sqrt{\frac{1}{2N}\log\frac{2}{\delta}},$$

and if instead h is chosen out of a **finite** set $\mathcal{H}$ of M hypotheses by looking at that same sample, the union bound gives, simultaneously for every $h \in \mathcal{H}$,

$$\left|E_{\text{in}}(h) - E_{\text{out}}(h)\right| \;\leq\; \sqrt{\frac{1}{2N}\log\frac{2M}{\delta}}.$$

Shatter coefficient and VC dimension. For binary classification, a **dichotomy** of N points $\mathbf{x}_1, \dots, \mathbf{x}_N$ is a sequence of labels $\big(h(\mathbf{x}_1), \dots, h(\mathbf{x}_N)\big) \in \{+1, -1\}^N$ produced by some $h \in \mathcal{H}$. The **shatter coefficient** $S(\mathcal{H}, N)$ is the largest number of distinct dichotomies $\mathcal{H}$ can produce on N points, where we are free to place the points wherever is most favourable; always $S(\mathcal{H}, N) \leq 2^N$. When $S(\mathcal{H}, N) = 2^N$ we say $\mathcal{H}$ **shatters** N points, and the **VC dimension** $d_{\text{VC}}(\mathcal{H})$ is the largest N for which this happens.

Remember:

> to show $d_{\text{VC}} \geq k$: exhibit **one** set of k points that $\mathcal{H}$ can label in *all* 2^k ways;
>
> to show $d_{\text{VC}} < k+1$: show that **no** set of $k+1$ points can be labelled in all ways.

Question 1 — The VC dimension

Take $\mathcal{X} = \mathbb{R}$ for the first hypothesis set and $\mathcal{X} = \mathbb{R}^2$ for the second:

$$\mathcal{H}_{\text{int}} = \left\{ h_{a,b}(x) = \begin{cases} +1, & a \leq x \leq b \\ -1, & \text{otherwise} \end{cases} \;:\; a \leq b \right\} \qquad (\textbf{positive intervals})$$

$$\mathcal{H}_{\text{cvx}} = \left\{ h_C(\mathbf{x}) = \begin{cases} +1, & \mathbf{x} \in C \\ -1, & \mathbf{x} \notin C \end{cases} \;:\; C \subset \mathbb{R}^2 \text{ convex} \right\} \qquad (\textbf{convex sets})$$

A set C is **convex** if the segment joining any two of its points lies entirely inside C; discs, half-planes, triangles and polygons are convex, a star-shaped form is not.

(a) Compute $d_{\text{VC}}(\mathcal{H}_{\text{int}})$.

(b) Compute $d_{\text{VC}}(\mathcal{H}_{\text{cvx}})$. [*Hint*: place the N points on the perimeter of a circle and, given a labelling of them, consider the polygon whose vertices are the points labelled $+1$.]

(c) Compare the two VC dimensions. Which of the two hypothesis sets is PAC learnable and which is not? Relate your answer to Popper's demarcation criterion: what does a report of $E_{\text{in}}(\hat{h}) = 0$ mean for each of the two?

(d) **(Optional)** Compute d_{VC} for each of the following:

(i) **positive or negative rays**: the hypotheses that are $+1$ on $[a, \infty)$ *together with* those that are $+1$ on $(-\infty, a]$;

(ii) **positive or negative intervals**: $+1$ on $[a, b]$ and -1 elsewhere, *together with* -1 on $[a, b]$ and $+1$ elsewhere;

(iii) **two concentric spheres in** $\mathbb{R}^2$: the hypotheses that are $+1$ exactly when $a \leq \|\mathbf{x}\| \leq b$. [*Hint*: what does such a hypothesis do to the single number $r = \|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2}$?]

Solution.

(a) $d_{\text{VC}}(\mathcal{H}_{\text{int}}) = \mathbf{2}$.

Two points are shattered: for $x_1 < x_2$, take $[a, b]$ containing neither point, only x_1, only x_2, or both, giving all 4 dichotomies.

No three points are: for any $x_1 < x_2 < x_3$, the dichotomy $(+1, -1, +1)$ needs $a \leq x_1$ and $b \geq x_3$, but then $x_2 \in [a, b]$ is labelled $+1$ too.

(b) $d_{\text{VC}}(\mathcal{H}_{\text{cvx}}) = \boldsymbol{\infty}$.

Place N points on a circle and label them arbitrarily. The polygon C with the $+1$ points as vertices is convex, and contains no -1 point, since all N points lie on the circle while C lies inside it. So h_C realises the labelling, and every N is shattered.

(c) $d_{\text{VC}} = 2$ against $d_{\text{VC}} = \infty$. Since ERM is consistent for every distribution exactly when the VC dimension is finite, $\mathcal{H}_{\text{int}}$ is **PAC learnable** and $\mathcal{H}_{\text{cvx}}$ is **not**.

Three points labelled $(+1, -1, +1)$ refute $\mathcal{H}_{\text{int}}$, so it is falsifiable and $E_{\text{in}}(\hat{h}) = 0$ is evidence. If the data generating distribution generates only points in a circle, then no dataset refutes $\mathcal{H}_{\text{cvx}}$: $E_{\text{in}}(\hat{h}) = 0$ is guaranteed before the data is seen, so it carries no information. Since we do not know the data generating distribution, we cannot rule this case out, nor any of the other unfalsifiable ones.

(d) (i) $d_{\text{VC}} = \mathbf{2}$. Two points: the right-pointing rays give $(-1, -1)$, $(-1, +1)$, $(+1, +1)$ and a left-pointing one gives $(+1, -1)$. Three points: a ray changes label at most once along the line, so $(+1, -1, +1)$ is unreachable.

(ii) $d_{\text{VC}} = \mathbf{3}$. A positive interval makes one contiguous block $+1$, a negative one makes the two outer blocks $+1$. Three points: $(+1, -1, +1)$ is the negative interval around x_2, the rest are positive intervals. Four points: $(+1, -1, +1, -1)$ needs $\{x_1, x_3\}$ to be one block or a prefix plus a suffix, and it is neither.

(iii) $d_{\text{VC}} = \mathbf{2}$. The hypothesis sees $\mathbf{x}$ only through $r = \|\mathbf{x}\|$, labelling $+1$ when $a \leq r \leq b$: a positive interval on the radius, so (a) applies.

Question 2 — Two ways of estimating E_{out}

A dataset has 600 examples. Before any fitting is done, a randomly chosen subset of 200 examples is **set aside as a test sample** and is never touched during learning. A hypothesis set with $M = 1\,000$ hypotheses is used, and the final hypothesis $\hat{h}$ is selected by ERM on the remaining 400 **training** examples. We wish to estimate $E_{\text{out}}(\hat{h})$, and we have two numbers available:

$E_{\text{in}}(\hat{h})$, the error of $\hat{h}$ on the 400 training examples;

$E_{\text{test}}(\hat{h})$, the error of $\hat{h}$ on the 200 test examples.

Take $\delta = 0.05$ throughout, so that all statements below hold with probability at least 95%. You may use $\log 40 = 3.689$ and $\log 40\,000 = 10.597$.

(a) Write down the error bar attached to $E_{\text{test}}(\hat{h})$ and compute it. Which of the two bounds recalled at the start of this tutorial did you use, and why is it legitimate here even though $\hat{h}$ was *chosen* using data?

(b) Write down the error bar attached to $E_{\text{in}}(\hat{h})$ and compute it. Why must M appear here and not in (a)?

(c) Which of the two estimates has the **higher** error bar? Comment on the fact that the training estimate is computed from *twice* as many examples as the test estimate.

(d) **(Optional)** How large would the training set have to be for its error bar to match the one you obtained in (a)? Compare with the 600 examples available.

Solution.

(a) Plain Hoeffding, with 200 examples and no M:

$$\left|E_{\text{test}}(\hat{h}) - E_{\text{out}}(\hat{h})\right| \;\le\; \sqrt{\frac{1}{2(200)}\log\frac{2}{0.05}} = \sqrt{\frac{3.689}{400}} = \mathbf{0.096}.$$

It is legitimate because the test examples took no part in producing $\hat{h}$, so relative to them $\hat{h}$ is a fixed hypothesis.

(b) The union bound, since $\hat{h}$ is the winner of a search over $M = 1\,000$ hypotheses on those same examples:

$$\left|E_{\text{in}}(\hat{h}) - E_{\text{out}}(\hat{h})\right| \;\le\; \sqrt{\frac{1}{2(400)}\log\frac{2(1000)}{0.05}} = \sqrt{\frac{10.597}{800}} = \mathbf{0.115}.$$

M appears because $E_{\text{in}}(\hat{h})$ is the smallest of 1 000 numbers computed on those examples.

(c) The **training** estimate, 0.115 against 0.096, despite twice as many examples: doubling the sample size divides the bar by $\sqrt{2}$, while the search multiplies the quantity under the root by $\log(2M/\delta)/\log(2/\delta) \approx 2.9$.

(d) Solving $\sqrt{\log(2M/\delta)/(2N)} = 0.096$ for N,

$$N \;=\; \frac{\log(2M/\delta)}{2(0.096)^2} \;=\; \frac{10.597}{0.018444} \;\approx\; \mathbf{575}.$$

Out of the 600 available, this leaves only 25 for testing, whose error bar would then be $\sqrt{3.689/50} = \mathbf{0.271}$, far worse than the 0.096 we started from.

Question 3 **How to run a tipping scam (and how not to fall for one)**

The scheme. Scammers have a list of 64 000 email addresses. On the Monday before each of 5 consecutive Canberra Raiders NRL matches, they email a tip (**win** or **lose**) to everyone still on their list, choosing what to send to each recipient by tossing a virtual fair coin, **independently** of the other recipients and of the other matches. After the match they keep on their list only those recipients to whom they happened to send the *correct* tip, and drop the rest. After the fifth match they email everyone left on the list, offering the sixth tip for a payment of **\$500** and advertising that they have tipped the correct result five matches in a row. Sending one email costs them \$0.01.

(a) Over the 5 matches, a recipient receives a sequence of 5 tips. How many different such sequences are there, and how many of the original 64 000 recipients are expected to receive each one? How many recipients are therefore expected to end the five weeks having been sent five correct tips in a row?

(b) How many emails do the scammers expect to send over the five weeks, and what do they cost them?

(c) What proportion of the recipients who received five correct tips must pay the \$500 for the scam to turn a profit? Would you have guessed that number before computing it?

(d) You are one of the recipients holding a perfect record of five correct tips. Set the situation up in the language of this course: what plays the role of the sample, of a hypothesis (what are the features and output, and where do they take values), and of the hypothesis set $\mathcal{H}$? What is E_{in} of the tipster you have been watching? Have your IML lectures saved you from falling for a scam?

Solution.

(a) Each match admits 2 tips, so over 5 matches there are $2^5 = \mathbf{32}$ possible sequences of tips, each received by a given recipient with probability $2^{-5} = 1/32$. Writing $Z_i = \mathbb{1}\{\text{recipient } i \text{ receives that sequence}\}$ and using linearity of expectation,

$$\mathbb{E}\Big[\sum_{i=1}^{64\,000} Z_i\Big] = \frac{64\,000}{32} = \mathbf{2\,000}.$$

One of the 32 sequences is the sequence of true results, so **2 000** recipients are expected to end the five weeks having received five correct tips in a row. Equivalently, the list halves in expectation each week: $64\,000 \to 32\,000 \to 16\,000 \to 8\,000 \to 4\,000 \to 2\,000$.

(b) By (a) the expected list sizes are $64\,000, 32\,000, 16\,000, 8\,000, 4\,000$, so **124 000** emails are expected, costing $124\,000 \times \$0.01 = \mathbf{\$1\,240}$, plus 2 000 emails for the sales pitch, a further \$20.

(c) At \$500 each they need $\$1\,260/\$500 = 2.52$ payers, that is 3 people out of the 2 000 with a perfect record:

$$\frac{3}{2\,000} = \mathbf{0.15\%}.$$

(d) The **sample** is the five matches, so $N = 5$. The **features** identify the match (teams, venue, date), taking values in some set $\mathcal{X}$; the **output** is the result, $\mathcal{Y} = \{\text{win}, \text{lose}\}$. A

hypothesis is a tipster $h : \mathcal{X} \to \mathcal{Y}$, and the **hypothesis set** $\mathcal{H}$ is the collection of all 32 sequences the scammers sent out, so $M = 32$. The tipster you watched made no mistakes, so $E_{\text{in}} = 0$.

The 32 sequences are *every* labelling of the five matches, so $\mathcal{H}$ shatters the sample: whatever the results had been, some hypothesis of $\mathcal{H}$ was bound to have $E_{\text{in}} = 0$, and the scammers showed you exactly that one. A zero in-sample error that was, by design, certain in advance carries no information about E_{out}.

Question 4 **The fish, the net, and assumptions that fail silently**

To determine the distribution of the sizes of fish in a lake, a net is cast and the fish caught are measured. The fractions of fish of each size in the catch are computed, and, the sample being large, statistical conclusions are drawn about the sizes of the fish in the entire lake.

Recall the assumption under which every guarantee of this course was derived: the sample $\mathcal{D}_N = \{(\mathbf{X}_1, Y_1), \ldots, (\mathbf{X}_N, Y_N)\}$ consists of independent draws from the same distribution P that generates the future examples on which E_{out} is defined.

(a) Can you smell a problem with the fish sample? Describe the mechanism precisely, name the distribution the catch is actually a sample from (call it P') and say how it differs from P.

(b) Does catching ten times as many fish in this way repair the problem?

(c) Give an analogous failure mode in the bank's credit problem of Lecture 1. [*Hint*: for which applicants does the bank get to observe whether they repay?]

(d) What could be done, about the fish and about the bank? Give one fix that changes the **data collection** and one that changes the **claim being made**.

Solution.

(a) The net has a **mesh**: fish smaller than it swim through and are never measured. The catch is an i.i.d. sample not from P, the distribution of sizes in the lake, but from P', namely P conditioned on the fish being caught, which over-represents large fish. This is **sampling bias**. What fails is not independence, and not that the fish are identically distributed, but that $P' \neq P$.

(b) No. The error bar $\sqrt{\frac{1}{2N} \log \frac{2}{\delta}}$ shrinks towards zero, so the estimate converges to the average under P': more data narrows the interval around the wrong centre.

(c) The bank observes repayment only for the applicants it **approved in the past**, so its data is a sample of applicants accepted under the rule then in force, not of applicants in general. The rule is then deployed on all applicants, including the region of $\mathcal{X}$ its data never contained.

(d) *Data collection*: use nets of several mesh sizes; for the bank, approve a small random fraction of the applicants who would otherwise be rejected and record their outcomes.

Claim: state the conclusion for the population actually sampled: "among fish larger than 10 cm…", "among applicants accepted under the current rule…".